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YouTube Trending Video Analysis Dashboard

Problem Statement

YouTube generates a large amount of video data daily, making it difficult to analyze which videos are trending and what type of content performs better. Understanding user engagement through views, likes, and comments across different regions is also challenging.

The objective of this project is to analyze YouTube trending video data and build an interactive dashboard that provides insights into video performance, category distribution, and engagement patterns across multiple regions.

Solution / Features

This project provides a Power BI dashboard that helps in analyzing YouTube trending videos in an easy and interactive way.

Key features of the solution include:

- Data from multiple regions (India, US, Korea, Russia) is combined into a single dataset
- A new column “Region” is created to differentiate data based on location
- Data cleaning and transformation is done using Power Query
- A separate category mapping table is created to convert category IDs into readable category names
- Relationship is established between tables for proper filtering

The dashboard includes:

- KPI cards showing total views, likes, comments, and total number of videos
- Bar chart displaying top 10 videos based on views
- Pie chart showing category-wise distribution of views
- Line chart representing trend of views over time
- Scatter plot showing relationship between views and likes
- Region slicer to filter data dynamically

Screenshots

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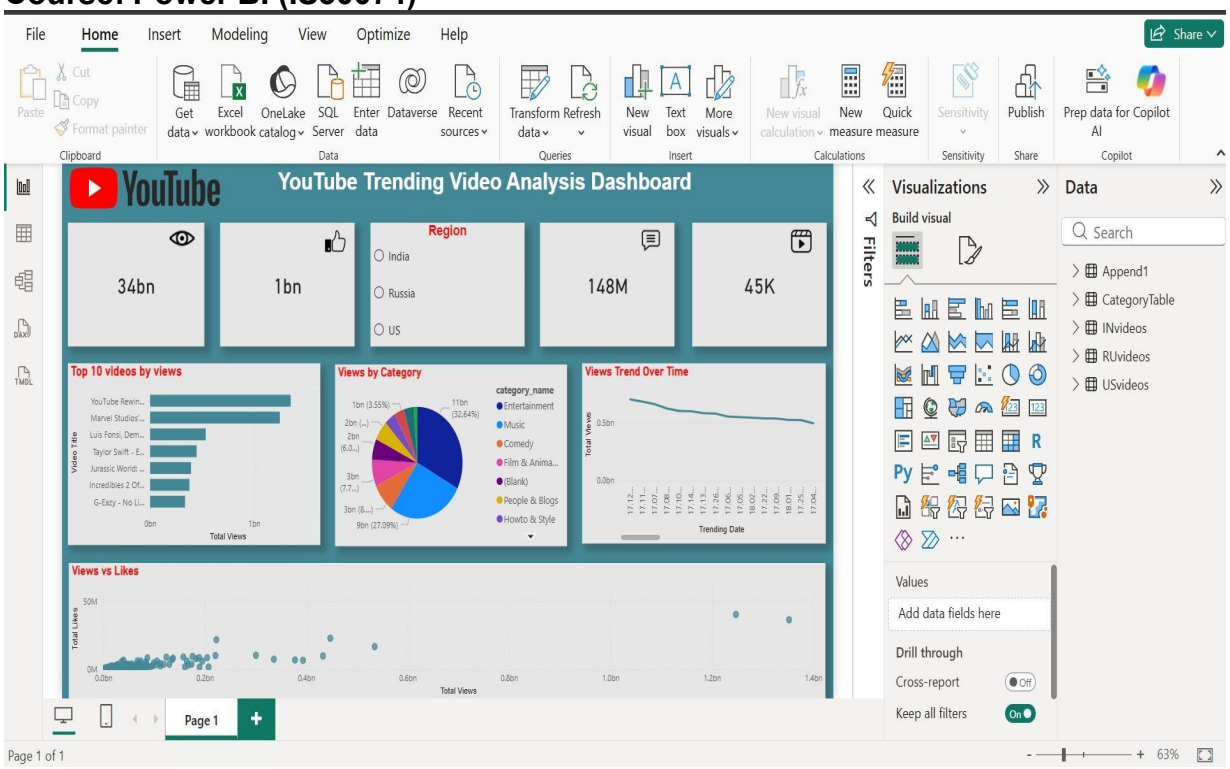

Tech Stack

The following tools and technologies were used to successfully complete this project:

- **Power BI – for creating dashboard and visualizations:**
  Power BI is used as the main tool to design and develop the interactive dashboard. It helps in creating different types of visualizations such as bar charts, pie charts, line charts, and scatter plots. It also enables the creation of KPI cards and interactive slicers, making the dashboard dynamic and user-friendly.
- **Power Query – for data transformation and preprocessing:**
  Power Query is used for cleaning and preparing the data before visualization. Tasks such as importing multiple datasets, adding a new "Region" column, combining datasets using append queries, and organizing the data were performed using Power Query.
- **CSV datasets – for input data:**
  The project uses CSV files containing YouTube trending video data from different regions. These datasets include information such as video title, views, likes, comments, category ID, and trending date, which are used for analysis.
- **DAX (Data Analysis Expressions) – for basic aggregations:** DAX is used in Power BI for performing calculations and aggregations. In this project, basic aggregations such as total views, total likes, and total comments are used in KPI cards and visualizations.

Unique Points

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This project includes several unique features that make it more effective and insightful compared to basic dashboards.

- **Multi-region analysis using combined dataset**
  Data from multiple regions such as India, United States, Korea, and Russia has been combined into a single dataset. This allows comparison of trends and performance across different regions within a single dashboard, making the analysis more comprehensive.
- **Interactive dashboard with slicer for region filtering**
  A region-based slicer is implemented, which allows users to dynamically filter the dashboard based on selected regions. This makes the dashboard interactive and enables users to analyze data specific to a particular region without creating multiple dashboards.
- **Use of category mapping to improve readability**
  Instead of displaying numerical category IDs, a separate mapping table is created to convert category IDs into meaningful category names such as Music, Entertainment, and Sports. This improves readability and makes the dashboard more user-friendly.
- **Combination of different types of charts for better analysis**
  Multiple visualization types such as bar charts, pie charts, line charts, and scatter plots are used together. Each chart provides a different perspective of the data, helping users understand performance, trends, and relationships more effectively.
- **Clean and structured dashboard layout**
  The dashboard is designed with a clear structure, including KPI cards at the top, followed by charts arranged in an organized manner. Proper alignment, spacing, and labeling are used to ensure that the dashboard is easy to understand and visually appealing.

Future Improvements

- The current dashboard provides meaningful insights into YouTube trending videos, but it can be further enhanced with additional features and advanced analytics.
- **Integration with real-time YouTube data using APIs**
  Currently, the dashboard is based on static datasets. In future, it can be improved by integrating real-time data using YouTube APIs. This will allow automatic updates of trending videos and provide up-to-date insights without manual data upload.
- **Adding more regions and datasets**
  The current analysis includes a limited number of regions such as India, US, Korea, and Russia. The project can be expanded by including more countries and datasets, which will provide a broader understanding of global trends and audience preferences.
- **Including sentiment analysis of comments**
  At present, the dashboard focuses on quantitative metrics like views and likes. In future, sentiment analysis can be performed on video comments to understand user opinions and emotions, which will add a qualitative aspect to the analysis.

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- **Enhancing dashboard design and user interaction**
  The dashboard design can be further improved by adding advanced UI elements such as buttons, tooltips, dynamic titles, and drill-through features. This will make the dashboard more interactive and user-friendly.
- **Adding predictive analysis features**
  Advanced analytics techniques such as forecasting and predictive modeling can be applied to predict future trends of views and engagement. This will help in identifying potential trending content and making data-driven decisions.

Conclusion

This project successfully demonstrates how Power BI can be used to analyze and visualize large datasets efficiently. It provides valuable insights into YouTube trending videos, helping users understand content performance and trends across different regions.